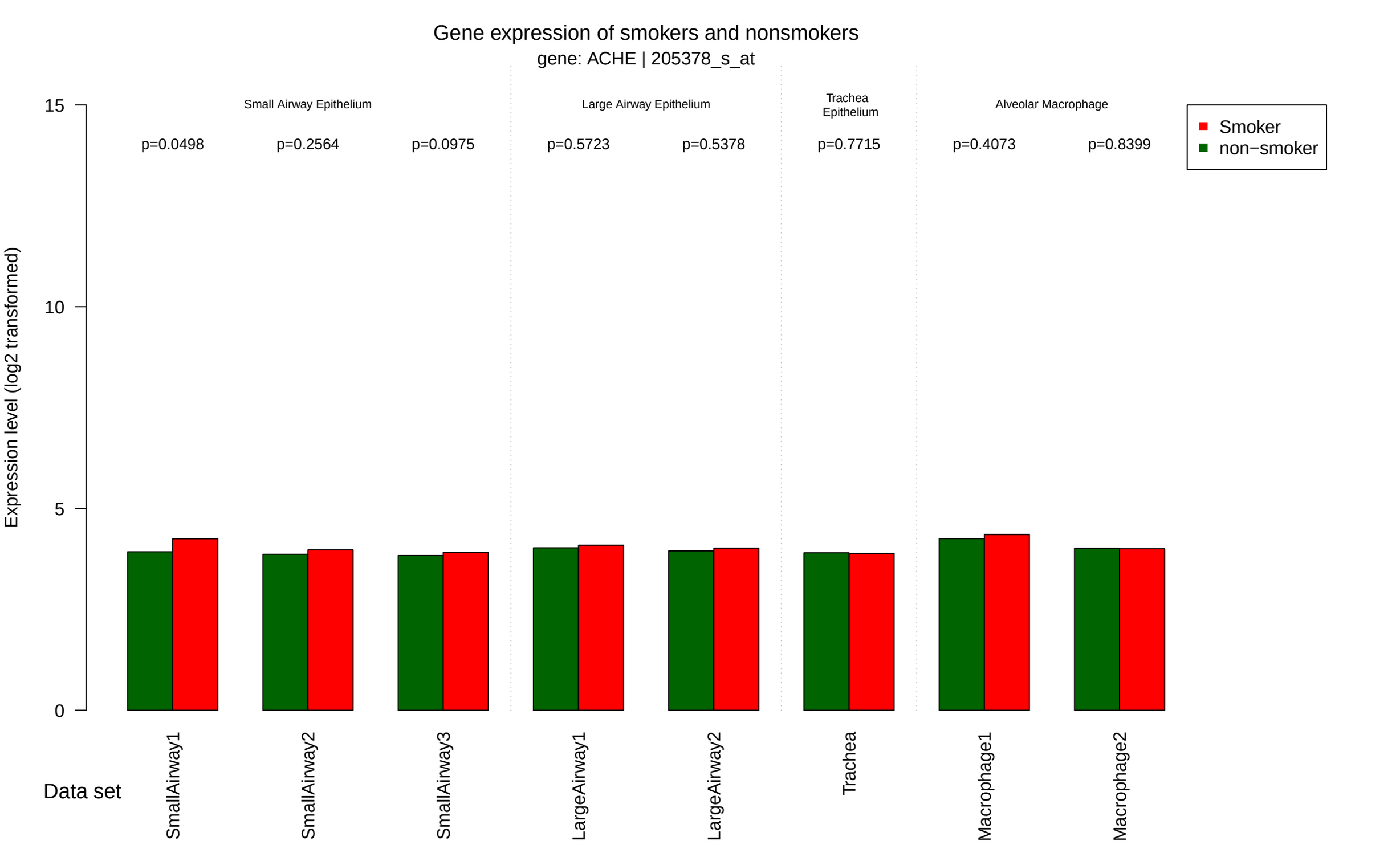
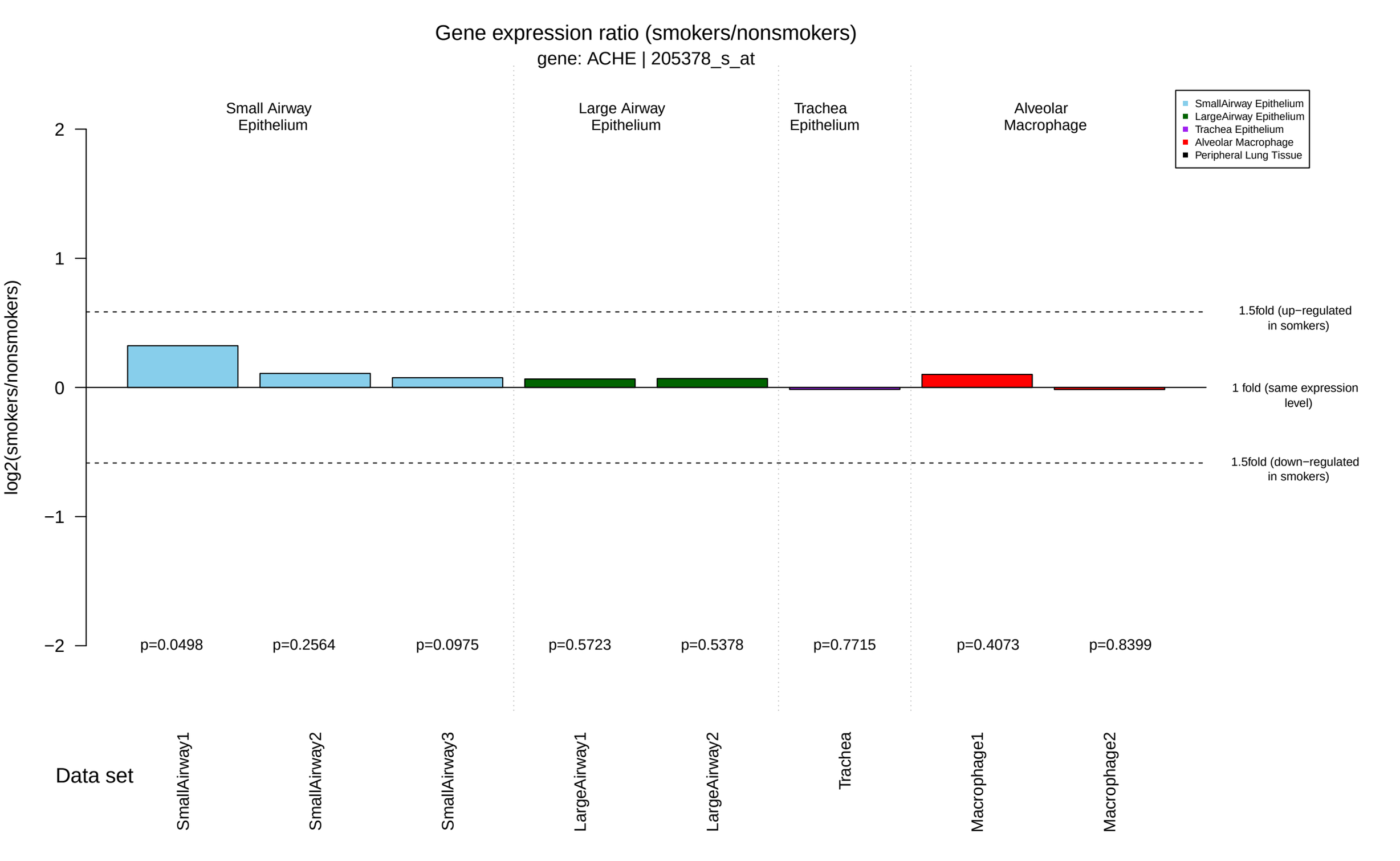
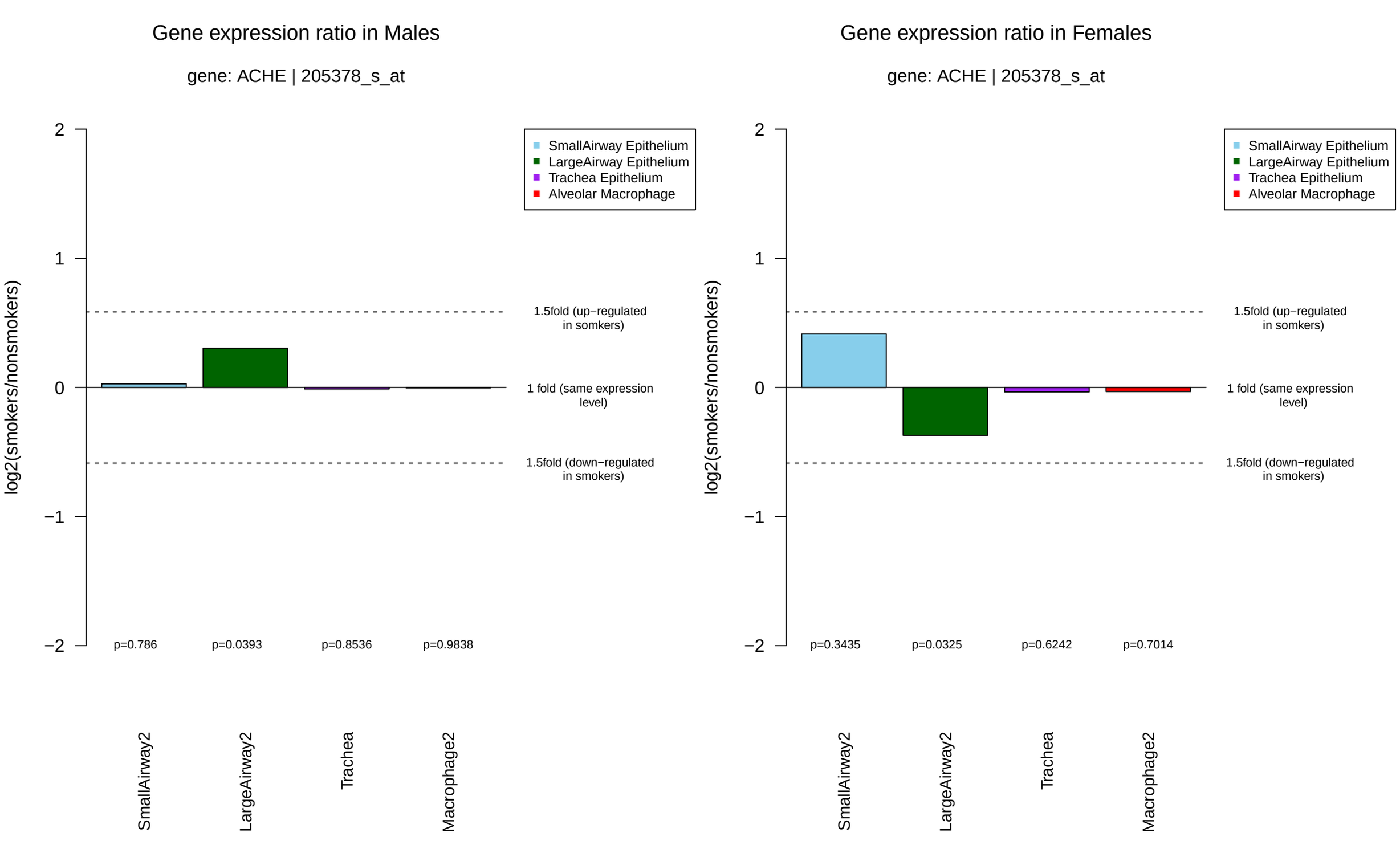
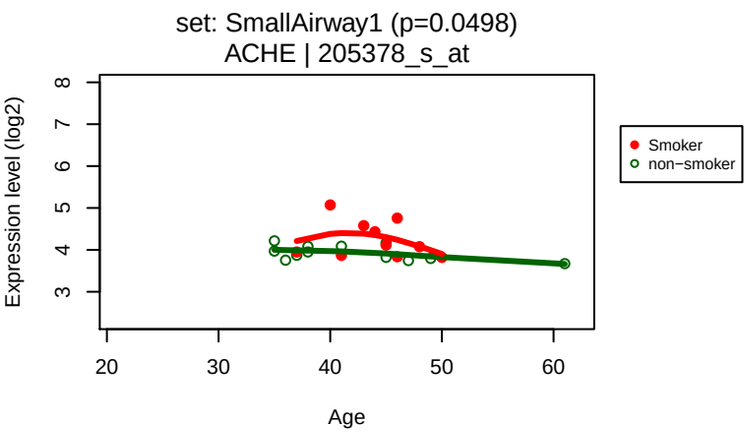
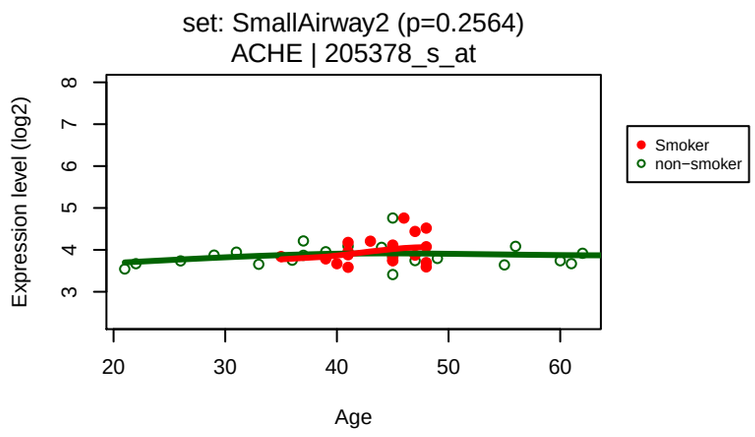
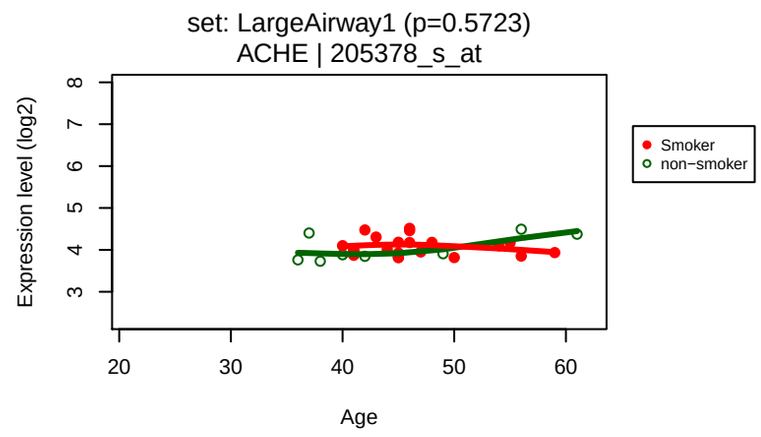

Gene expression of smokers and nonsmokers
gene: ACHE | 205378_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0498
p=0.2564
p=0.0975
p=0.5723
p=0.5378
p=0.7715
p=0.4073
p=0.8399
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

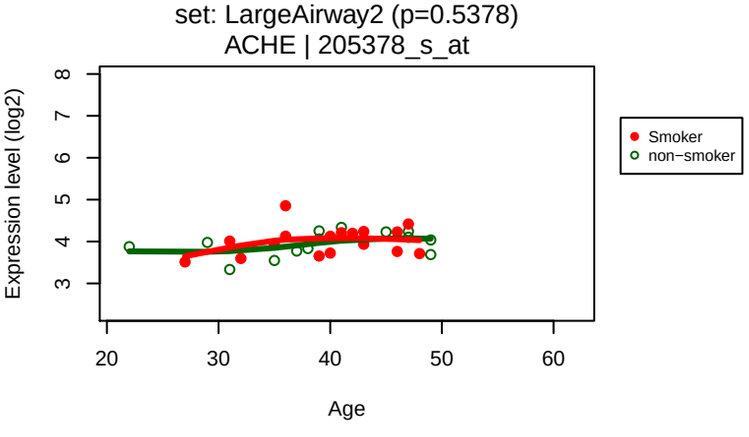

Gene expression ratio (smokers/nonsmokers)
gene: ACHE | 205378_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0498
p=0.2564
p=0.0975
p=0.5723
p=0.5378
p=0.7715
p=0.4073
p=0.8399
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

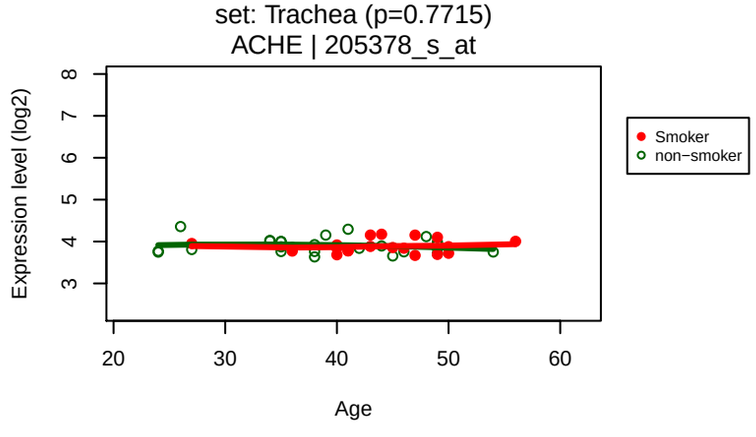

Gene expression ratio in Males
gene: ACHE | 205378_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.786
p=0.0393
p=0.8536
p=0.9838
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACHE | 205378_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3435
p=0.0325
p=0.6242
p=0.7014
SmallAirway2
LargeAirway2
Trachea
Macrophage2

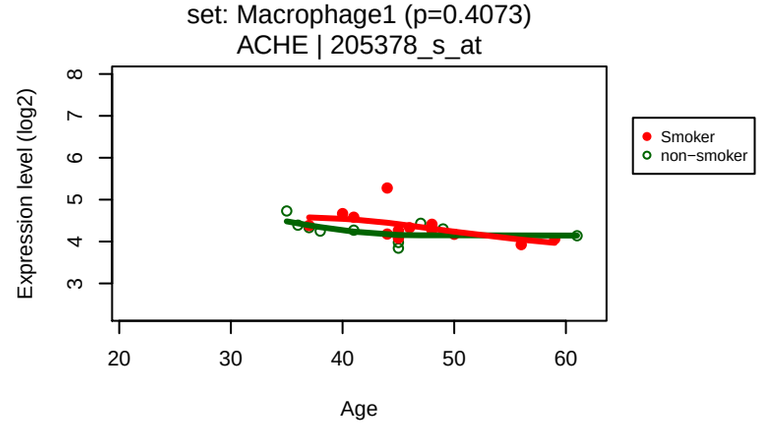

set: SmallAirway1 (p=0.0498)
ACHE | 205378_s_at
Expression level (log2)
Age
Smoker
non−smoker

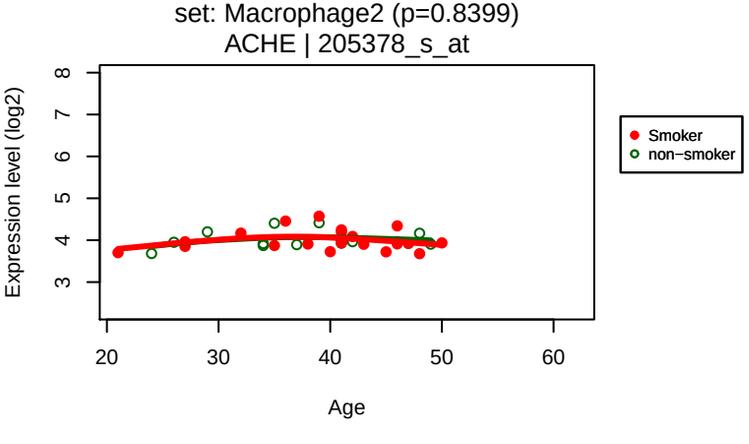

set: SmallAirway2 (p=0.2564)
ACHE | 205378_s_at
Expression level (log2)
Age
Smoker
non−smoker

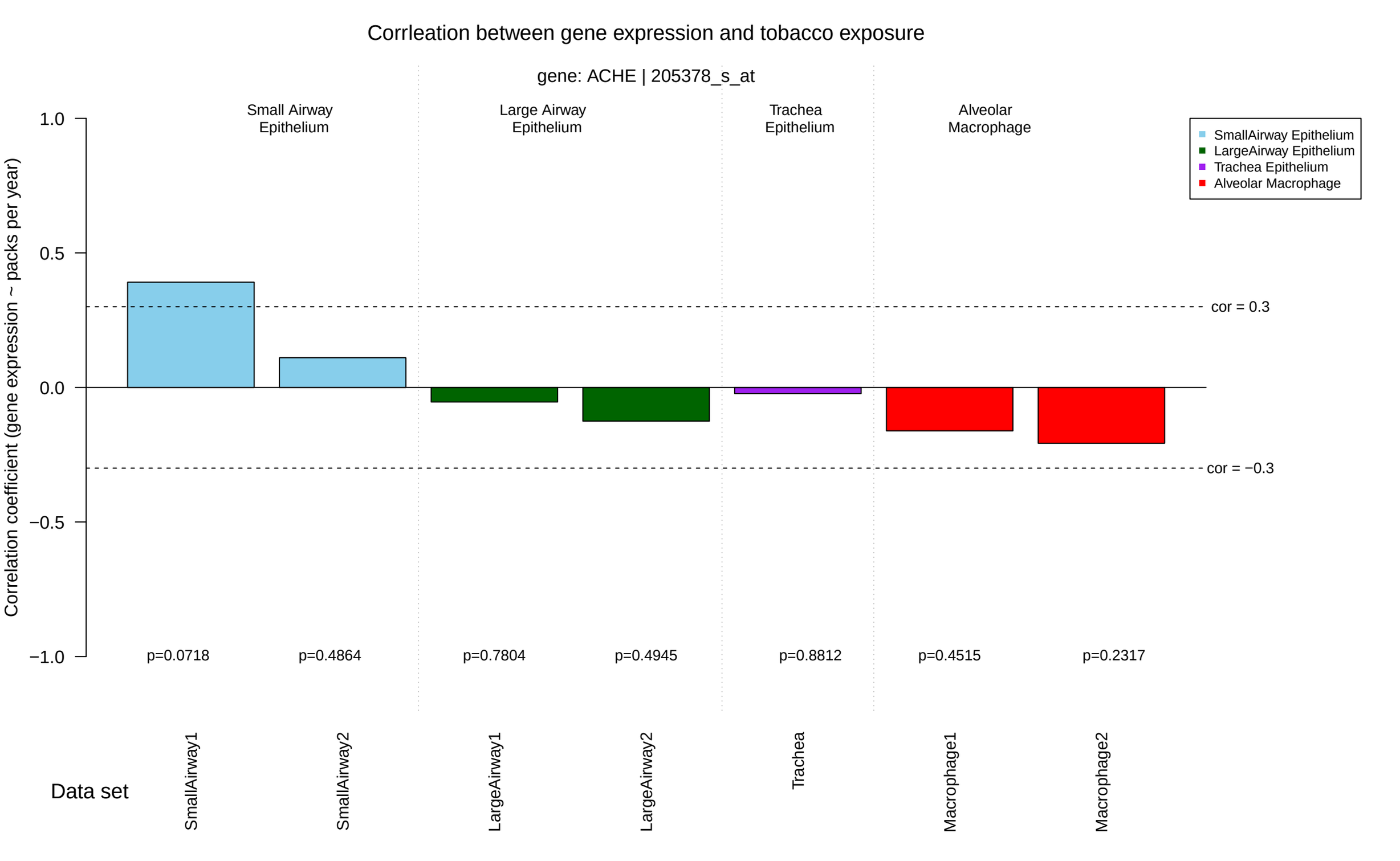

set: LargeAirway1 (p=0.5723)
ACHE | 205378_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5378)
ACHE | 205378_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7715)
ACHE | 205378_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4073)
ACHE | 205378_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8399)
ACHE | 205378_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACHE | 205378_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0718
p=0.4864
p=0.7804
p=0.4945
p=0.8812
p=0.4515
p=0.2317
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2